

# AeroMorse Usage Guide — Changing Your Device Safely

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This guide is for the day-to-day job of **changing an AeroMorse that is already working**: adding a macro, remapping a pattern, changing a password or a setting. It explains what each helper tool does and gives a tested, step-by-step order that keeps a working device working.

**Why the care?** For many AeroMorse users the device is their *only* way to use a computer. A single bad character in `morse_map.py` can stop it from starting. Every step below exists so that can't happen — and so that if it ever does, you can put the old files back in a minute.

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## The five rules

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1. **Back up the device before every change.** One double-click, dated, see [Back up my AeroMorse](#).
  2. **Never edit files directly on the device.** Edit the copies in your work folder, check them, then copy them over.
  3. **Copy nothing to the device until the validator says PASS.**
  4. **Save as UTF-8, not "UTF-8-BOM"** (Notepad++: *Encoding* → *UTF-8*). A BOM stops the device; the validator catches it.
  5. **Keep the last good backup until the new version has worked for a day.**
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## 1. Set up your work folder (one time)

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Create a folder: `Documents\AeroMorse`. This is your *work folder* — everything you edit and every tool you use lives here, and your backups go in a `Backups` folder inside it.

## What goes in it

| Put in the work folder                                                                                                                        | From                                     | Why                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>boot.py</code> , <code>code.py</code> , <code>config.py</code> , <code>morse_map.py</code>                                              | <b>your device</b> (the CIRCUITPY drive) | The copies you edit. Take them from the device, not the repo, so your own settings and patterns are what you edit. |
| <code>macro_secrets.enc</code> (or <code>macro_secrets.txt</code> )                                                                           | your device, if you already have one     | Your passwords (see <a href="#">AeroMorse Secrets</a> ).                                                           |
| <code>macro_secrets.example.txt</code>                                                                                                        | the repo                                 | Template for your <b>first</b> secrets file — see <a href="#">First time</a> .                                     |
| <code>aeromorse_validator.exe</code> + <code>Check my AeroMorse files.bat</code>                                                              | the <a href="#">AeroMorse repo</a>       | Safety check before copying to the device.                                                                         |
| <code>AeroMorse Secrets.exe</code> + <code>Edit my AeroMorse secrets.bat</code>                                                               | the repo                                 | Edit and PIN-lock your passwords.                                                                                  |
| <code>Back up my AeroMorse.bat</code>                                                                                                         | the repo                                 | One-click dated backup of the device.                                                                              |
| <code>morse_map_analyzer.exe</code>                                                                                                           | the repo                                 | Finds duplicate and free patterns.                                                                                 |
| <code>aeromorse_cheatsheet.htm</code>                                                                                                         | the repo                                 | Interactive cheat sheet of <i>your</i> map.                                                                        |
| <code>AeroMorse Cheat Sheet.pdf</code> , <code>AeroMorse – Keycode Reference.pdf</code> , <code>AEROMORSE_BUILD_GUIDE.pdf</code> , this guide | the repo                                 | Reference to read or print.                                                                                        |

When finished it looks like this:

```
Documents\AeroMorse\  
  boot.py  code.py  config.py  morse_map.py  macro_secrets.enc  
Back up my AeroMorse.bat  
Check my AeroMorse files.bat      aeromorse_validator.exe  
Edit my AeroMorse secrets.bat     AeroMorse Secrets.exe  
morse_map_analyzer.exe           aeromorse_cheatsheet.htm  
AeroMorse Cheat Sheet.pdf        (and the other PDFs)  
Backups\  
  AeroMorse-Green-2026-09-26_1430\  
  ...
```

## Give each device a name

Put an empty text file on the device named `AeroMorse-<Name>.txt` — e.g. `AeroMorse-Green.txt` . It does nothing on the device, but the backup tool and AeroMorse Secrets use it to label that device, which matters

as soon as you have two.

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## 2. The tools — what each one does

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### Back up my AeroMorse

**Back up my AeroMorse.bat** — double-click with the device plugged in. It copies *everything* on every AeroMorse drive it finds into `Backups\<name>-<date>_<time>\` (for example `Backups\AeroMorse-Green-2026-09-26_1430`). It only reads the device; nothing there is changed. Two devices plugged in = two backups in one click.

**Use it:** at the start of every change, before a CircuitPython update, and any time the device is working well and you want a known-good copy.

### Validator — "Check my AeroMorse files"

**Check my AeroMorse files.bat** runs `aeromorse_validator.exe` on the files in the work folder and answers one question: *will these files load on the device, or lock me out?*

- Checks `boot.py`, `code.py`, `config.py`, `morse_map.py` and your secrets file for the things that stop the device: a BOM, a Python syntax error, an import error, letter-O/letter-I typed instead of zero/one in a pattern, a bad setting value.
- **PASS** (green) = safe to copy. **FAIL** (red) = fix the line it names first. **WARN** (yellow) = the device will still run, but look at it (for example a secret key with no value, or a pattern stored under the wrong length so it can never be used).
- **Hidden characters:** if a stray invisible character slipped in while typing (for example from Ctrl+E), it lists each one and asks "Remove them now? Type Y then Enter". It saves a backup ( `<file>.before-fix-<time>.bak` ) before it changes anything.
- It never prints a password — only key names and counts.

### AeroMorse Secrets

**Edit my AeroMorse secrets.bat** opens `AeroMorse Secrets.exe` on the work folder. Passwords and personal details never go in `morse_map.py`; they live in a separate secrets file, and `morse_map.py` only *names* them. A line like

```
g3[4][0b0110] = _secret('password1', '(set password1 in macro_secrets.txt)') # ... P
```

means: *pattern* `...` in Group 3 types whatever is stored under the key `password1` — or the placeholder text in brackets if there's no such key.

There are two kinds of secrets file:

| File                           | What it is                                                                        | On the device                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <code>macro_secrets.txt</code> | Plain text, <code>key=value</code> lines. Anyone who opens the drive can read it. | Always available, no PIN.                                                     |
| <code>macro_secrets.enc</code> | The same lines, <b>encrypted</b> by AeroMorse Secrets with a PIN you choose.      | Locked at every power-up until you type the PIN in Morse. <b>Recommended.</b> |

## First time: creating your secrets

You start from the template, `macro_secrets.example.txt`.

- Copy the template.** In your work folder, copy `macro_secrets.example.txt` and rename the copy to `macro_secrets.txt`. (Windows may hide the `.txt` ending — if the copy is shown as `macro_secrets.example - Copy`, rename it to just `macro_secrets`.)
- See which keys your map uses.** Open `morse_map.py` and search for `_secret(`. Each one names a key — in the standard map: `name`, `address`, `phone`, `email`, `password1`, `wifi` (Group 3 letters A B C E P W) and `bank_login` (Group 6 R). The template already has a line for each.
- Fill in your values.** Open `macro_secrets.txt` in Notepad++ and replace each example value with your real one — everything after the first `=`: `name=Jane Smith password1=My-Real-P@ssword` - One secret per line. No quotes needed; symbols such as `@ ! $ # =` are fine in a value. - The key (before `=`) must be spelled **exactly** as in `_secret('...')`. - Delete lines you don't need. Add a line for any key you add to `morse_map.py` later. - Lines starting with `#` are notes; you can delete them. - Save as **UTF-8** (Notepad++: *Encoding* → *UTF-8*).
- Encrypt it (recommended).** Double-click **Edit my AeroMorse secrets.bat**. AeroMorse Secrets opens and imports `macro_secrets.txt` automatically — the status line says *"Imported plain macro\_secrets.txt"*. - Check the lines look right. - Type a **PIN** in both boxes (tick **Show PIN** to see it). Use letters and digits you can type in Group 1; 8 or more is best; capitals don't matter. - Click **Save (encrypt)**. It tells you about any key your `morse_map.py` uses that still has no value, then offers to **delete the plain `macro_secrets.txt`** — say **Yes** (the encrypted file now holds everything).
- Put it on the device.** Click **Copy to device** and pick the device. It offers to remove any old plain-text secrets file from the device — say **Yes**. The device restarts; don't sip or puff for ~5 seconds.
- Try it.** Use a secret pattern (e.g. Group 3 `P`). The screen shows **ENTER PIN + ENTER**: type your PIN in Morse with the Group 1 letters/digits, then Enter ( `. - . -` ). It unlocks and types the secret.

*No PIN wanted?* Skip steps 4–5 and copy `macro_secrets.txt` itself to the device. It works the same, but anyone who plugs the device into a computer can read your passwords.

## Later: changing a secret

- Double-click **Edit my AeroMorse secrets.bat**. It asks for your PIN to open `macro_secrets.enc`.
- Change, add or delete `key=value` lines.
- Save (encrypt)** — leave the PIN boxes empty to keep your current PIN, or type a new one twice to change it.
- Copy to device** — to every device that uses these secrets.

Forgot the PIN? Only the secrets are affected; keyboard and mouse keep working. Start again from the template (First time, step 1).

## On the device

- Secrets start **locked** after every power-up. The first secret you use asks for the PIN; nothing you type while entering it reaches the computer.
- Group 3 **U** ( `.-.` ) unlocks without typing anything; **L** ( `.-..` ) locks again. Esc cancels PIN entry.
- The screen and USB log only ever show `SECRET <key>` , never the value.

## Cheat sheet

`aeromorse_cheatsheet.htm` — open it in your browser (double-click). If the patterns don't appear by themselves, click  **Select morse\_map.py** and pick the one in your work folder. It shows every pattern of *your* map, group by group, as dots and dashes; click a row to hear its timing;  **Print Cheat Sheets** makes a paper copy (or *Microsoft Print to PDF*). Secrets show only their key name with a  , never the value.

Optional: `Build my cheat sheet.bat` (needs Python and a copy of the repo) prints `AeroMorse Cheat Sheet.pdf` from your map in one click.

## Morse map analyzer

`morse_map_analyzer.exe` — double-click; it reads the `morse_map.py` next to it and writes `morse_map_report.txt` (open it in Notepad). The report lists:

1. **Duplicate patterns** in a group (the later one silently wins);
2. patterns that **clash with Group 0** (the always-on group-switch codes win);
3. **unused patterns** of each length in each group — where to put a new macro without disturbing anything.

## Reference documents

- **Keycode Reference** (PDF) — every `Keycode.NAME` you can use in `morse_map.py` , and how to type shifted symbols.
- **Build Guide** (PDF) — hardware, wiring, every `config.py` setting (§10) and troubleshooting (§12).
- **README** — group-by-group pattern tables and the full secrets guide.

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## 3. The recommended order for any change

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This is the order used when updating the AeroMorse devices this project was built on. Each step protects the next one.

| #  | Step                                                                       | Tool                                            | Why at this point                                                                                            |
|----|----------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1  | <b>Back up the device</b>                                                  | Back up my AeroMorse                            | A dated, known-good copy <i>before</i> anything changes.                                                     |
| 2  | <b>Refresh your work copy</b> of the file you'll edit from the device      | File Explorer                                   | So you edit what is really on the device, not an older copy. Skip if you're sure the work folder is current. |
| 3  | <b>Plan:</b> find a free pattern                                           | Analyzer report, Keycode Reference              | Avoids reusing a pattern that already does something.                                                        |
| 4  | <b>Edit</b> <code>morse_map.py</code> (or <code>config.py</code> )         | Notepad++ (UTF-8)                               | Make the change.                                                                                             |
| 5  | <b>Re-run the analyzer</b>                                                 | Analyzer                                        | Catches a duplicate or Group 0 clash you just created.                                                       |
| 6  | <b>Secrets</b> — only if you added or renamed a <code>_secret()</code> key | AeroMorse Secrets                               | Add the value, <b>Save (encrypt)</b> . Its key check reads the <code>morse_map.py</code> you just edited.    |
| 7  | <b>Validate</b> — must say <b>PASS</b>                                     | Check my AeroMorse files                        | The final gate: checks every file exactly as the device will load it. Fix and re-run until PASS.             |
| 8  | <b>Copy to the device</b> only the file(s) you changed                     | File Explorer (secrets: <i>Copy to device</i> ) | The device restarts by itself. <b>Don't sip or puff for ~5 seconds</b> while it recalibrates.                |
| 9  | <b>Test on the device</b>                                                  | —                                               | Try the new pattern; if you changed secrets, try one (PIN).                                                  |
| 10 | <b>Update the cheat sheet</b>                                              | Cheat sheet / Build my cheat sheet              | So the printed sheet matches the device.                                                                     |

If anything goes wrong after step 8, restore — see [Putting a backup back](#).

Why the validator is *after* Secrets and the analyzer: it is the last word on whether the files are safe, so it should see the finished files. Running it earlier is fine too — just always run it again last.

## 4. Changing `config.py`

Same order as above (skip steps 3, 5, 6 and 10). Things to know:

- Settings are explained in the file itself and in **Build Guide §10**.
- `config.py` is **per device**. Two devices can need different sensor settings (for example a different puff threshold). Before editing, copy *that* device's `config.py` into the work folder (step 2), and copy it back only to that device.
- On/off settings accept `True / False` in any case; don't put them in quotes.

## 5. Putting a backup back

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1. Open `Backups\` and pick the newest folder for that device from *before* the change (the name has the date and time).
2. Copy the file(s) you changed from that folder back onto the device — usually just `morse_map.py`, `config.py` or the secrets file. The device restarts by itself.
3. If the device won't start at all, copy **all** the `.py` files from the backup, then unplug and replug it.

There is no need to delete the broken version first — copying over it is enough.

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## 6. More than one device

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- Give each device its own label file ( `AeroMorse-Green.txt`, `AeroMorse-Blue.txt` ); backups and AeroMorse Secrets then show which is which.
  - Give each its own `DEVICE_NAME` in `config.py` (e.g. `"AeroMorse Green"` ), so the computer lists them by name instead of two identical "Feather ESP32-S3 Reverse TFT" entries. It applies after an unplug/replug. (If you leave it as the default "AeroMorse", the label file is used instead. A wireless display board has no `config.py` ; give it a label file such as `AeroMorse-Display.txt` and it shows as "AeroMorse Display".)
  - A **wireless display** keeps its own settings in `receiver_config.py` on the display board (channel, brightness, rotation, timeouts). Its `ESPNOW_CHANNEL` must match the main device's `config.py` . Back it up and edit it the same way as the main device's files.
  - `code.py` and `boot.py` should be the **same version** on every device. `morse_map.py` can be shared if you want the same patterns everywhere. `config.py` usually differs.
  - One `macro_secrets.enc` (and one PIN) can go on every device — *Copy to device* asks which one.
  - The four files carry a version line near the top ( `AeroMorse code.py – version 1.13 ...` ); keep all four at the same version.
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## 7. One-page checklist

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- [ ] 1 Back up my AeroMorse (dated backup made)
- [ ] 2 Work copy refreshed from the device
- [ ] 3 Free pattern found (analyzer report)
- [ ] 4 Edited in Notepad++ - saved as UTF-8
- [ ] 5 Analyzer re-run - no new duplicates / Group 0 clashes
- [ ] 6 New secret keys? -> AeroMorse Secrets -> Save (encrypt)
- [ ] 7 Check my AeroMorse files -> PASS
- [ ] 8 Copied changed file(s) to the device - no sip/puff for 5 s
- [ ] 9 Tested on the device
- [ ] 10 Cheat sheet updated